

DSA 0601 Data handling Visulazation

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Set 11: Product Category Analysis

Data Visualization using R

Aim

To analyse product category sales using a funnel chart, a table, a pie chart and a dashboard in R.

Procedure

1. Open RStudio and load the packages ggplot2 and gridExtra (install them once with `install.packages(c("ggplot2", "gridExtra"))`).
2. Create the sales data as a data frame and sort it from highest to lowest sales.
3. Question 1: draw the funnel chart with `geom_tile()`; the stages are the categories ranked by sales.
4. Question 2: make the table with `tableGrob()`.
5. Question 4: draw the pie chart with `geom_col()` and `coord_polar("y")`.
6. Question 3: join the pie chart, funnel chart and table into a dashboard with `arrangeGrob()`.
7. Run the code and note the output and the figures.

R Code

```
library(ggplot2); library(gridExtra)

# Dataset
sales <- data.frame(Category = c("Electronics", "Clothing", "Appliances"),
  Sales = c(50000, 35000, 40000))
sales$Percent <- round(sales$Sales / sum(sales$Sales) * 100, 1)
sales <- sales[order(-sales$Sales), ]
print(sales)
sales$Category <- factor(sales$Category, levels = rev(sales$Category))

# 1. Funnel chart
p1 <- ggplot(sales, aes(0, Category)) +
  geom_tile(aes(width = Sales, fill = Category), height = 0.8) +
  geom_text(aes(label = paste0(Category, ": $", Sales))) +
  labs(title = "Sales Funnel by Category") + theme_void()

# 2. Table
tb <- tableGrob(sales, rows = NULL)

# 4. Pie chart
p4 <- ggplot(sales, aes("", Sales, fill = Category)) + geom_col() + coord_polar("y") +
  geom_text(aes(label = paste0(Percent, "%")), position = position_stack(vjust = 0.5)) +
  labs(title = "Sales Distribution") + theme_void()

# 3. Dashboard
dash <- arrangeGrob(p4, p1, tb, layout_matrix = rbind(c(1,2), c(3,3)), top = "Product Category Dashboard")

ggsave("set11_q1.png", p1, width = 6, height = 4, bg = "white")
ggsave("set11_q2.png", arrangeGrob(tb, top = "Sales by Category"), width = 5, height = 2, bg = "white")
ggsave("set11_q3.png", dash, width = 12, height = 8, bg = "white")
ggsave("set11_q4.png", p4, width = 6, height = 4, bg = "white")
```

Output

	Category	Sales	Percent
1	Electronics	50000	40
3	Appliances	40000	32
2	Clothing	35000	28

Sales Funnel by Category

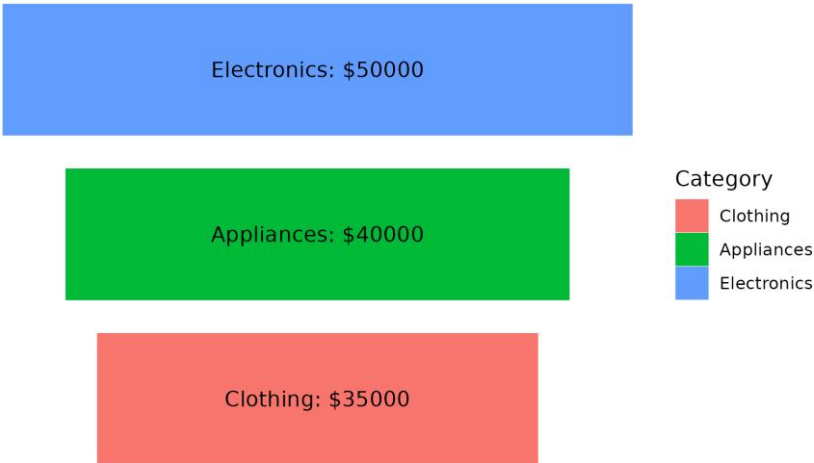


Figure 11.1: Q1 - Funnel chart

Sales by Category

Category	Sales	Percent
Electronics	50000	40
Appliances	40000	32
Clothing	35000	28

Figure 11.2: Q2 - Sales table

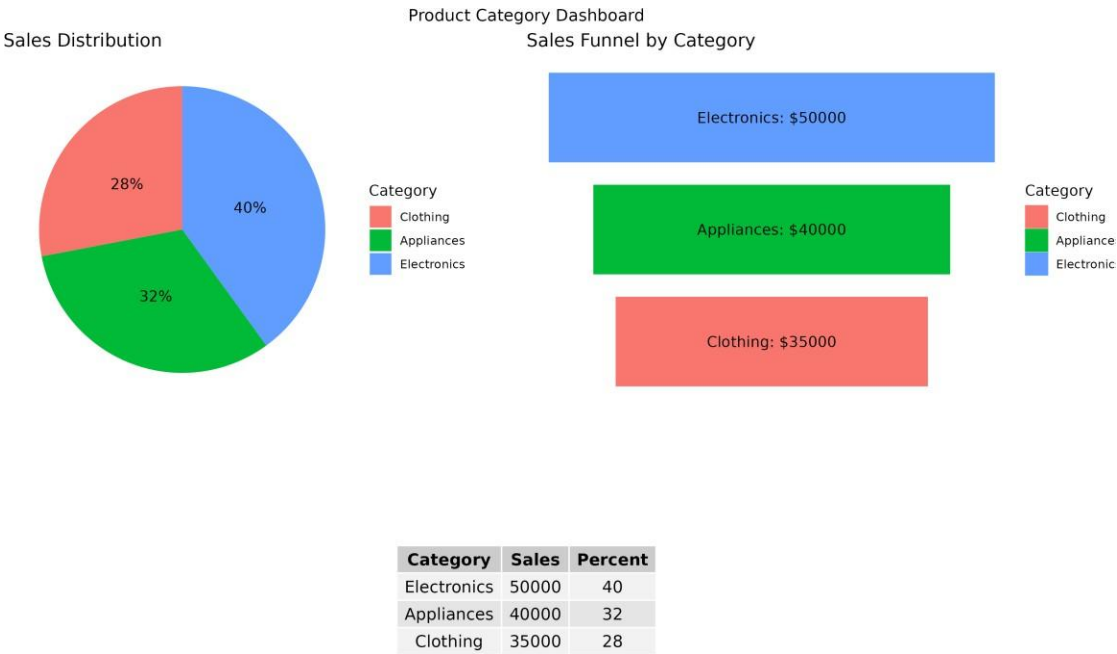


Figure 11.3: Q3 - Dashboard

Sales Distribution

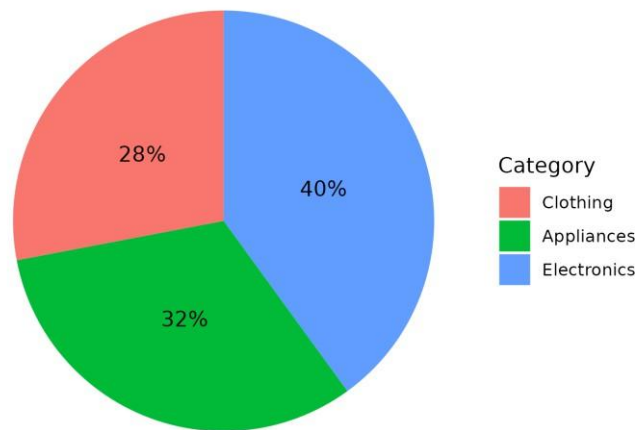


Figure 11.4: Q4 - Pie chart

Result

- Total sales are \$125,000.
- Electronics has the highest share (40%), then Appliances (32%) and Clothing (28%).
- The data has no conversion stages, so the funnel shows categories ranked by sale

